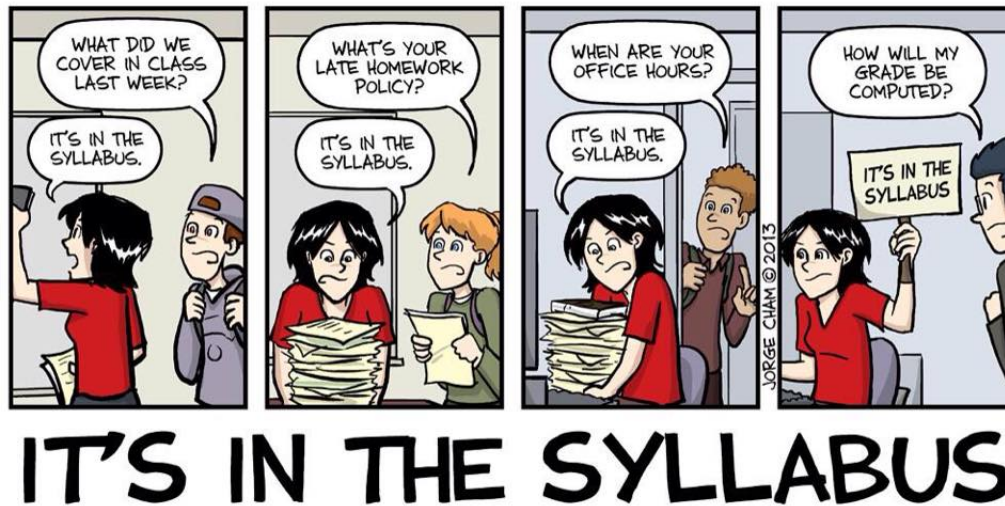


CSC4910 CS Capstone (Fall 2025)



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WWW.PHDCOMICS.COM

INSTRUCTOR:	Christopher Stuetzle
E-MAIL:	stuetzlec@merrimack.edu
OFFICE:	
LECTURE:	TR 2:00PM–3:45PM in 510 Turnpike Room 105
OFFICE HOURS:	Available By <i>Google Calendar</i> Appointment
WEBSITE:	Google Classroom
REQUIRED TEXTBOOKS:	None

OFFICIAL COURSE DESCRIPTION

This CS Major capstone course is an in-depth study of the methodologies involved in researching, designing, developing, implementing, and maintaining software products. The course will focus on large-scale semester-long projects with multiple developers, during which students work in small development teams producing a product for a client. Students put to use good object-oriented design practices and methods. This course serves as a capstone experience calling on a variety of knowledge and skills and involving significant practice in professional writing in both collaborative and individual environments for different audiences: the team, the manager, the client, and the user, with an emphasis on professional conduct, skills, and both written and oral communication. *Four hours per week.*

PREREQUISITES

CSC 2620 with a C or better, and senior standing as a CS major

COURSE OBJECTIVES

The primary objectives of this course are to

1. To master the tasks of requirements elicitation and documentation, in order to understand and document the needs of a client.
2. To master the tasks of functional specification documentation.
3. To collaboratively create and execute a long-term project plan for the implementation of a large-scale project including a timeline with milestones and the appropriate delegation of tasks within a team.
4. To execute formal communication with client, superior, teammates, and end-users in both oral and written forms, including writing support and use documentation.

EXPECTATIONS

This is your capstone course, and as such it carries the following expectations:

- I expect that you will commit **considerable** time to this course outside of class. What “considerable” means is dependent upon your current point and role within your semester-long project, but expect to spend more time on this course work than other courses you’ve had.
- I expect that you attend all lectures and meetings, and that you make your schedule as flexible as possible to accommodate the schedules and needs of your teammates.
- I expect that you “do your part” at all stages of the project development and that you keep up with the assigned reading and writing assignments.
- I expect you to be courteous and professional when dealing with each other, with myself, and most importantly with your client.

Activity	Approximate Hours
Reading/Reviewing/Studying	1
Working on Semester Project	7-9
Class Meetings	2-4

I further expect students to keep up with the corresponding reading and watching. Remember, you will only get out of this class what you put into it.

LECTURE AND ATTENDANCE

I expect students to attend all lectures, pay full attention, and take notes when appropriate. You should read ahead in the textbook so you are prepared for each class. I will be posting lecture notes for each class ahead of time on Google Classroom. ***You will need to take notes in addition to these.*** General good practice is to download and print these documents to take additional notes on them, or simply take notes in a notebook and use the lecture notes as reference.

Please respect your classmates by silencing your cell phones and other electronic devices before class begins. I will be doing the same. There will be **no text messaging, instant messaging, gaming, or surfing the web** during class. No electronic devices of any kind (phones, calculators, laptops, etc.) are allowed during lecture. We will do occasional in-class exercises, during which you may use personal laptops. If you are disruptive in class I will ask you not to be. If you continue to be disruptive, I will ask you to leave.

Important Note: This class will be following the Department of Computer and Data Sciences' Attendance Policy: We will take attendance every class. Excessive lateness counts as an absence. If a student misses more than 2 weeks' worth of class meetings (4 in total), they will receive an automatic F in the class.

COMMUNICATION

I will be communicating through e-mail as needed. This will be the primary method for dissemination of information between us, so please check your e-mail at least once a day. I check my e-mail several times a day, so it is the quickest way to reach me with questions (aside from visiting office hours). I will also be archiving important announcements through Google Classroom.

When e-mailing me, include *a greeting and a closing signature*, or I will likely not respond. Using my name as a greeting and yours as a closing signature is sufficient.

If you are having trouble with an assignment or an upcoming quiz, please do not suffer in silence. Let me know if you are struggling, either in my office hours (during which you can just stop by) or by scheduling an appointment. I'm interested in seeing students succeed and am more than happy to provide extra help.

If you are a non-native speaker of English, translation devices are encouraged during class and quizzes. During these quizzes, however, no mobile devices capable of accessing the Internet (particularly smartphones) are allowed, so you will need to use a translator that cannot access the Internet.

ACADEMIC INTEGRITY

All written work for this class must be your original work. Presenting material from other sources, either print or electronic, as one's own work constitutes plagiarism. Please review Merrimack College's Academic Integrity Code: <http://catalog.merrimack.edu/content.php?catoid=9&navoid=202#academic-integrity> Please consult the library's web site for a complete discussion on academic integrity <http://libguides.merrimack.edu/content.php?pid=120821> for a complete discussion of academic integrity.

A note about LLMs: You are free to consult LLMs for ideation, outlining, and basic/fundamental code writing. But you are responsible for its output and you are responsible for producing a quality

product for your client. If LLM usage enhancing your personal workflow, I encourage its usage but remember you are producing a polished, bug-free (as much as possible) product.

GRADING

If you wish to dispute a grade, you must do so **within one week** of receiving the grade. This is non-negotiable. Your final grade will be determined according to the rubric on the Semester Project document.

All grading will be on a single point scale. At the end of the semester, I will simply add the number of points you've earned, divide by the number of points possible (295), weighted according to your group contribution. Your grade will be heavily affected by your individual contribution to your group, as demonstrated through *issues closed*.

Your semester grade breaks down as follows:

Activity	Points Possible	Approximate Portion of Final Grade
Semester Project	265	87%
Contribution/Miscellaneous Points	40	13%

ACADEMIC ACCOMMODATIONS FROM ACCESSIBILITY SERVICES OFFICE

Regardless of whether the course or the student is on-campus or remote, Merrimack College provides reasonable accommodations for students with documented disabilities through the Accessibility Services Office. Students who have, or think they may have, a disability are invited to contact the Accessibility Services Office via the online request form found on the Accessibility Services website: www.merrimack.edu/aso, email accessibilityservices@merrimack.edu or by visiting us on the third floor of McQuade Library (subject to change if the college is remote).

Students are encouraged to contact the office as soon as possible via the website or via email at accessibilityservices@merrimack.edu to ensure adequate time to meet and create a plan. Students already registered with Accessibility Services are encouraged semesterly to request for their letters to be emailed and students are responsible to then email the letter to their instructors personally. The Accessibility Services Testing Center remains available to students whether in-person or remote. While it is understood that some students will not use all accommodations in all courses, accommodations can not be made retroactively.

LEARNING CONTINUITY AND CONTINGENCY PLANS

In the event of a snow day, we will still meet using Google Meet linked from Google Classroom. Our meeting will be casual, and will count towards attendance. If we lose power, we will not meet.

MERRIMACK COLLEGE WITHDRAWAL POLICY

Please be mindful of the academic calendar and what the last day to drop, add, and withdraw from the class are. The Registrar's Office publishes and manages those deadlines. Please be aware that a form submitted to the instructor just before the Registrar's deadline does not satisfy the need to get any signed forms to the Registrar's Office by their published deadlines.

ALL ARE WELCOME

This classroom is a safe and collaborative environment and diversity will be respected including differences in race, religion, class, sexual orientation, gender and gender-identity, age, body-type, disability, and psychological struggle. Because we live in a global world, members of this class have diverse language backgrounds. While oral communication in a new language takes only a few years, academic language acquisition takes 5-7 years. Like those of us with other differences, students with accents will be respected in this class. If there is anything that I can do to make your time in the class better, please do not hesitate to let me know.

MENTAL HEALTH

It is important to make your mental health a priority! If you want more information or resources, please come see me! If you are struggling with your mental health, or you believe a classmate is, please talk to me so I can put you in touch with qualified and caring support to get you back on track and feeling better.

Suicide Prevention Lifeline

We can all help prevent suicide. The Lifeline provides 24/7, free and confidential support for people in distress, prevention and crisis resources for you or your loved ones, and best practices for professionals. http://www.ulifeline.org/stay_well National Suicide Prevention Lifeline 1-800-273-8255(TALK) or Text START to 741-741. Student Success Resources It takes a village to progress in your knowledge. The below resources are a sampling of what Merrimack College provides to assist you in meeting your goals. If you have questions, please see me for additional information.

Academic Success Center The Academic Success Center offers workshops, programs, tutoring, and individual or group meetings to all students. Topics include: academic expectations, dealing with test anxiety, effective reading, preparing for tests, study skills, and time management. Visit <https://www.merrimack.edu/academics/academic-success-center/>

RESOURCES ON CAMPUS

The Writing Center

Because everyone is a writer, the Merrimack College Writing Center provides free consultations for any student, in any major, in any class. Individual or small group consultations are generally done face-to-face in the Writing Center, McQuade Library 3rd floor. Limited online consultations are

also available. Visit www.merrimack.edu/writingcenter or visit MyMack for a listing of hours and for registration and appointment instructions. Please note that while drop-ins are welcome, appointments take priority. For specific information about availability, schedule, and location of all tutoring available on campus, please go to <http://www.merrimack.edu/tutoring>.

The Tutoring & Math Center

The Tutoring & Math Center operates both face-to-face and on Blackboard. Students may meet with tutors in a socially-distant, college, state, and CDC approved one-to-one face-to-face session by appointment. For virtual, drop-in assistance, students may log into Blackboard where a peer tutor will assist them in real-time via a Collaborate session – no appointment needed. All students in STEM courses have been given access to the Tutoring & Math Center Blackboard page, titled "Tutoring & Math Center Online Tutoring". Visit <https://www.merrimack.edu/academics/academic-success-center/tutoring-and-math-center> or email MathCenter@Merrimack.edu for more information.

McQuade Library

For help citing sources consult the Writing Center or try a citation management tool such as Mendeley or Zotero. McQuade Library LibGuides can help: <http://libguides.merrimack.edu/mendeley> <http://libguides.merrimack.edu/zotero>

Tech Support

If you are having trouble logging in to your accounts, or with your iPad, please contact the help desk by emailing askit@merrimack.edu. Here are some basics to get you started: https://www.merrimack.edu/about/offices_services/information-technology-services/students/

IN CLOSING

I am glad you are here! If you have questions or concerns, please reach out to me via email, phone, zoom, office hours, or after class.

SCHEDULE (SUBJECT TO CHANGE)

Day	Unit	Topics	Project Phase	Deliverable Due	Sprint
9/2	1	Syllabus, Project Setup	Setup		–
9/4	2	Agile Development Cycle, Kanban	Setup	Team Charter	1
9/9	3	Requirements Gathering	Requirements		1
9/11	4	Issue Tracking	Requirements		1
9/16	5	Test-Driven Design	Requirements	Requirements Draft	1
9/18	–	Report Outs, Work Day	Requirements		2
9/23	6	Functional Specification and Design	Design	Requirements Final	2
9/24	–	SECS Career Fair	Development	Attendance	2
9/25	7	User Interfaces, Design Patterns	Design		2
9/30	8	Profiling, Licensing	Design		2
10/2	–	Report Outs, Work Day	Design		3
10/7	PD1	Professional Development Day 1	Design	Functional Spec. Draft	3
10/9	–	Work Day	Design		3
10/14	–	FALL BREAK	–	–	3
10/16	–	Report Outs, Work Day	Development	Functional Spec. Final	4
10/21	–	Guest Lecture	Development		4
10/23	–	Work Day	Development		4
10/28	PD2	Professional Development Day 2	Development		4
10/30	–	Report Outs, Retrospective 1	Development		5
11/4	–	Work day	Development		5
11/6	9	Posters and Presentations	Development		5
11/11	–	VETERAN'S DAY	Development, Poster		5
11/13	–	Report Outs, Code Review	Development, Poster		6
11/18	10	User Manuals, Tutorials, Tech Docs	Development, Poster		6
11/20	–	Work Day	Development, Poster		6
11/25	–	Work Day, Report Outs	Development, Manual	Poster Draft	6
11/27	–	THANKSGIVING BREAK	Development, Manual		7
12/2	–	Work Day	Development, Manual	Report-Out, Manual Draft	7
12/4	–	Work Day, Presentation Practice	Development, Pres.	Poster Final	7
12/9	–	Report Outs, Retrospective 2	Development, Pres.	User Manual Final Tech. Document Draft	7
12/18	–	Presentation and Deployment		Software Deployed Tech. Manual Final	